

Brown, Nicholas

From: Spierings, H.T. (Eric) <E.Spierings@umcutrecht.nl>
Sent: Thursday, June 11, 2026 5:33 AM
To: Gragert, Loren; Brown, Nicholas
Subject: [External] RE: immunogenetr and new nomenclature

My comments below in red.

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Kind regards,

Eric Spierings

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Questions I would value your input on

1. In the tables you work with, how are serologic DQA / DPA / DP values actually written — a bare number, a prefixed token such as DQA1 or DPw4, or the leading-zero form such as 01?

Depends on if I created the table I'm working with. The UNOS/SRTR tables don't store the prefix, and I recalled NMDP's database systems did not store the prefix either, but that is not how I would design such a system. I'd prefer to have the full antigen names stored as strings in the database with the prefix like "DQA01", "DPA01", "DPB31", per our published PLString convention.

We follow ET guidelines. When executing the typings, we store DNA data in our labsystem and report that to our patient hospital system. Once we upload our HML, ET is doing the conversions. See below how that currently works. The DQA1 may be a good example for your case. Most important indicator (and the only official one) is the presence of an "*". Locus indicators are maintained.

Entry Date	TT Lab	Full Phenotype	Full Phenotype S
	NUTTT	A3, A3, B7, B40, B60, Cw3, Cw10, Cw7, DR6, DR13, DR7, DR52, DR53, DQ1, DQ6, DQ2, DQA-01, DQA-02, DP-03, DP-0401, DPA-01, DPA-01	N/A
CIWD Genotype		Full Phenotype	
A	A*03:01, A*03:01	A3, A3	
B	B*07:02, B*40:01	B7, B40, B60	
C	C*03:04, C*07:02	Cw3, Cw10, Cw7	
DRB1	DRB1*13:02, DRB1*07:01	DR6, DR13, DR7	
DRB345	DRB3*03:01, DRB4*01:01	DR52, DR53	
DQB1	DQB1*06:04, DQB1*02:02	DQ1, DQ6, DQ2	
DQA1	DQA1*01:02, DQA1*02:01	DQA-01, DQA-02	
DPB1	DPB1*03:01, DPB1*04:01	DP-03, DP-0401	
DPA1	DPA1*01:03, DPA1*01:03	DPA-01, DPA-01	
Publics			

2. Do your laboratories actually store associated antigens (the A0201 level of resolution) in these tables? Or is that level uncommon enough that it is reasonable to default ambiguous values to molecular and treat serologic as an explicitly declared case?

We aren't using associated antigens for reporting typing or antibody specificities in our clinical laboratory yet, except for DR103. Under OPTN Policies, an unacceptable antigen of DR1 does not screen off donors as incompatible who have a DR103 antigen for their typing, which is contrary to WHO nomenclature broad/split nomenclature rules. We report antibody specificities as a combination of split antigen categories and two-field molecular nomenclature. If we determined a typing of A*24:03:01:01 by NGS without any two-field ambiguity, on our reports we have it appear as an A24 split antigen instead of an A2403 associated antigen, even though "A2403" is what appears in the IPD-IMGT/HLA database as its "Pre-2026 WHO Assigned Serology".

<https://www.ebi.ac.uk/ipd/imgt/hla/alleles/allele/?accession=HLA00053>

For the typings, see above (so, No). For acceptables and unacceptable we follow the ET classifications and we are also able to communicate unacceptable antigens at the second field PL-string level. So in your case, we would put DRB1*01:03 as unacceptable. That will exclude DR103 while keeping DR1 open.

3. Is the DQA1-means-molecular versus DQA-means-serologic column convention something we can expect to take hold, or will column naming stay inconsistent for the foreseeable future?

I hope so and would prefer to always have separate columns for serologic versus molecular typing data, and at least on the backend and in electronic file formats use full WHO nomenclature to represent data in those columns as strings like "DRB1*01:03" and "DR0103". I'm always optimistic that consistency will improve.

I agree with Loren. Separate columns for molecular and serologic data are preferable, but the more important point is that the stored value itself should retain the nomenclature signal. For molecular/DNA typing, that means using full WHO allele nomenclature with the locus and `\*`, for example `DRB1\*01:03`, `DQA1\*01:02`, or `DPB1\*04:01`. For serologic data, I would also prefer full antigen strings such as `DR0103`, `DQA01`, or `DPB0201`, rather than bare numbers.

I would not rely too heavily on the `DQA1` versus `DQA` column-name convention. It is a good convention and should be encouraged, but I expect legacy datasets, registry exports, and manually curated tables to remain inconsistent. Column names can help as metadata, but they should not be the only basis for classification.

I would also be careful not to make `immunogenetr` responsible for cleaning up every possible malformed or lossy representation that people may create. That can easily become a Sisyphean task. The package can be robust and helpful, but it should also define a clear boundary: if the input has lost the information needed to distinguish molecular from serologic nomenclature, then the user needs to supply that information explicitly.

So I would frame the recommendation as: molecular values should include `\*`; serologic values should use explicit antigen nomenclature; and when that information has been stripped from the value, the user should be required to declare the column interpretation or accept an explicit warning/error. Once values have been reduced to bare strings such as `01` or `0201`, the ambiguity is real and software cannot reliably reconstruct the intended nomenclature from the value alone.

4. Do you have example tables — real or representative — that show the range of formats in use? Concrete examples would let me build a test set that prevents regressions and confirms the approach across the cases that actually occur.

See attached for a mapping table I developed for the CPRA calculator that has both antigen and molecular nomenclature. Once OPTN Policies supports more associated antigens besides DR103 (now renamed DR0103), this table will be updated using similar conventions. You can see I use the prefixes in the nomenclature. This table was created back in 2022. The genotype table itself also used two-field WHO molecular nomenclature with the prefixes - I think you got a data use agreement from NMDP and obtained that?

For the current Eurotransplant setting, the ETRL HLA-DQA1 table is a useful example of an operational mapping between molecular DQA1 allele nomenclature and ET match determinants. It lists molecular values such as DQA1*01:01 and corresponding determinants such as DQA-01, and is explicitly described as being used for HLA typing and unacceptable antigen listing in Eurotransplant. This is therefore a good example of a current, real-world transplant-system format.

However, I would treat this as an example of the current ET implementation, not as a final model for the new WHO nomenclature. Once the new nomenclature is implemented in Eurotransplant, this mapping will likely need to be updated or reinterpreted. It is still useful as a test case, because it shows that operational systems may use their own determinant labels alongside formal molecular allele names.

Loren

Eric

From: Brown, Nicholas <Nicholas.Brown@Pennmedicine.upenn.edu>

Date: Wednesday, June 10, 2026 at 10:40 AM

To: Spierings, H.T. (Eric) <e.spierings@umcutrecht.nl>; Gragert, Loren <lgragert@tulane.edu>

Subject: immunogenetr and new nomenclature

External Sender. Be aware of links, attachments and requests.

Hi Eric and Loren,

I sat down to update my immunogenetr package to appropriately recognize the new serologic nomenclature, only to find that it was more complicated than I expected. I've laid out my thoughts and some questions in the attached document. Would you mind reviewing and giving me feedback on my planned approach?

Thanks,

Nick

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